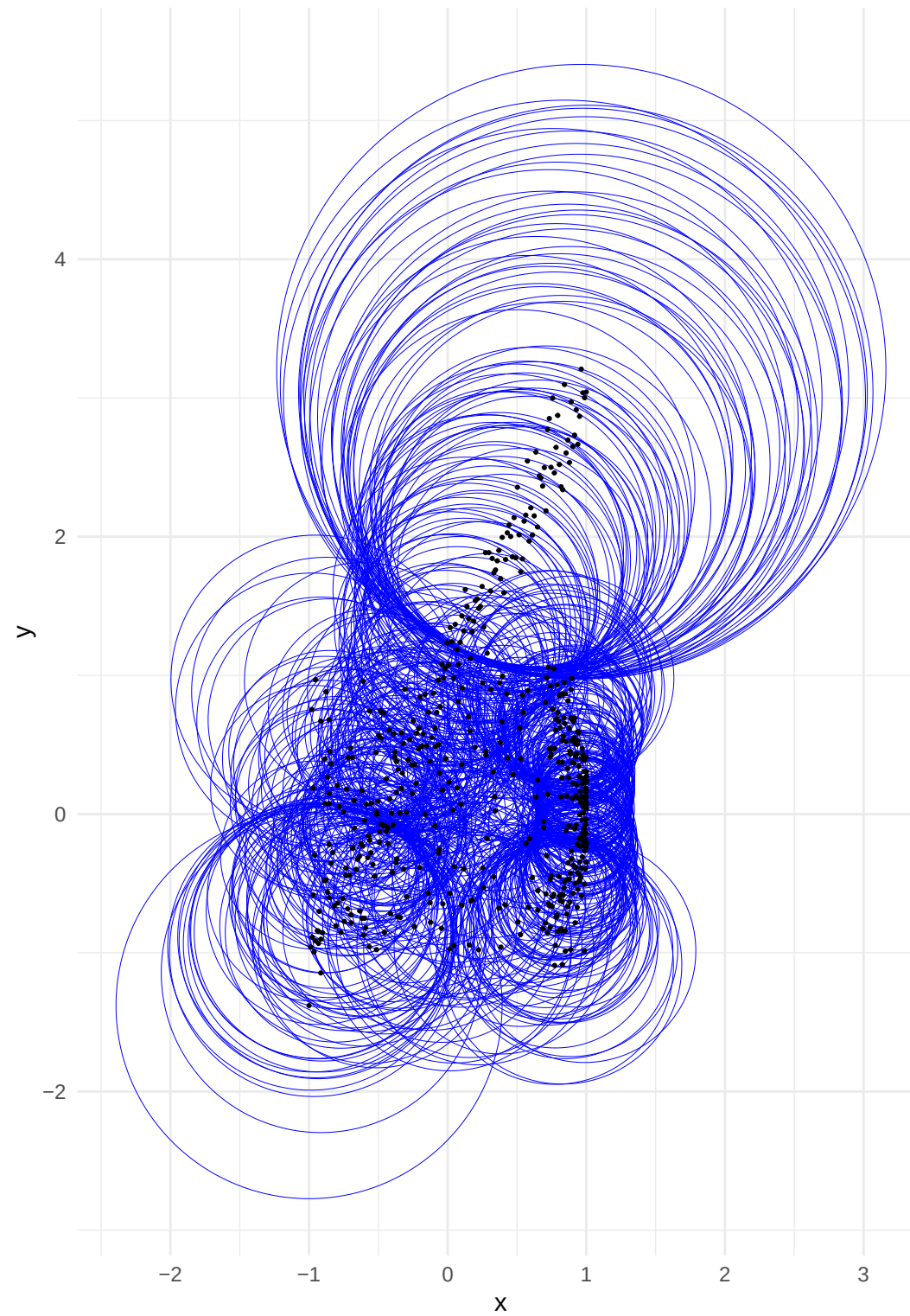
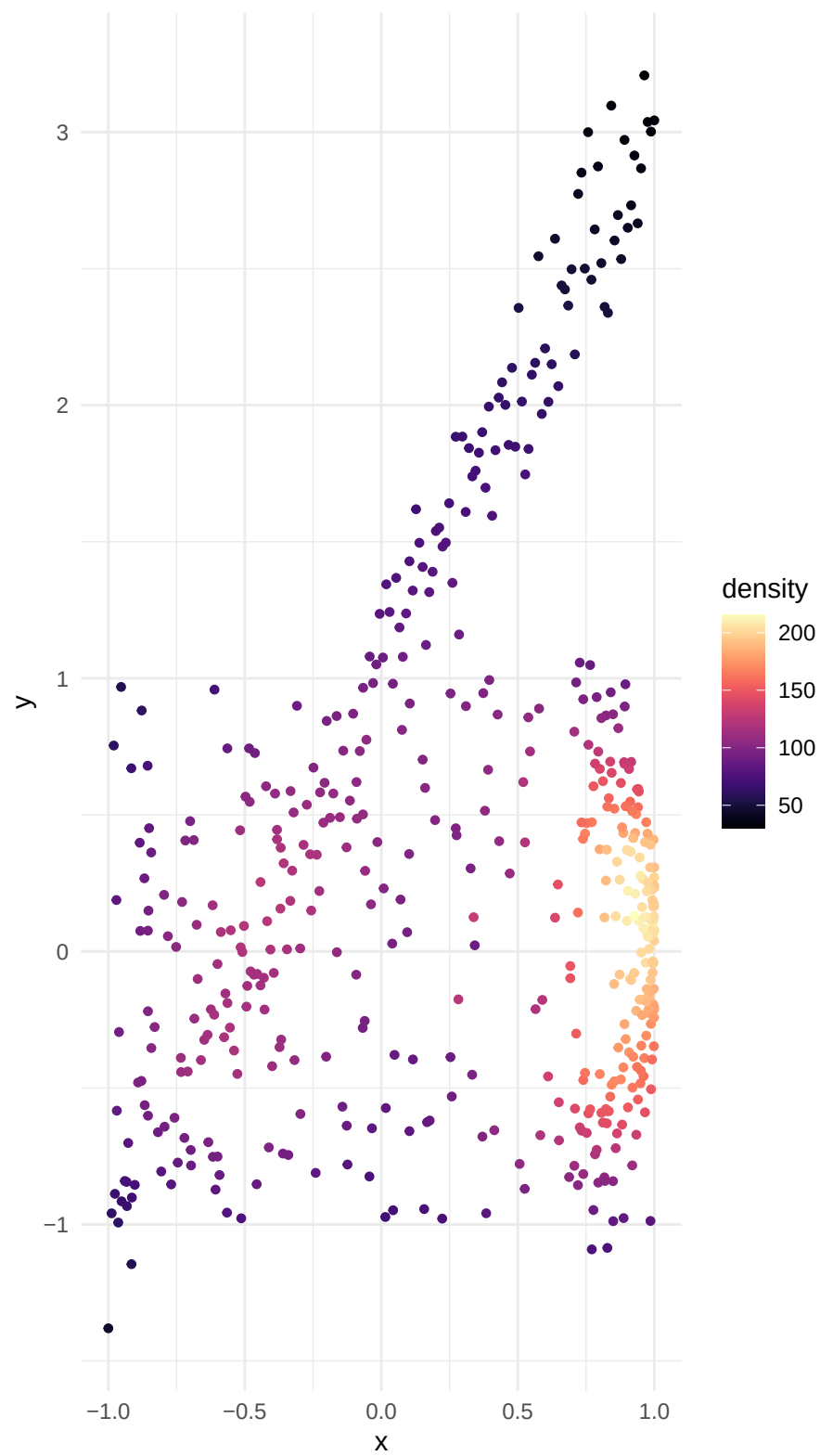


# All Adaptive Spheres

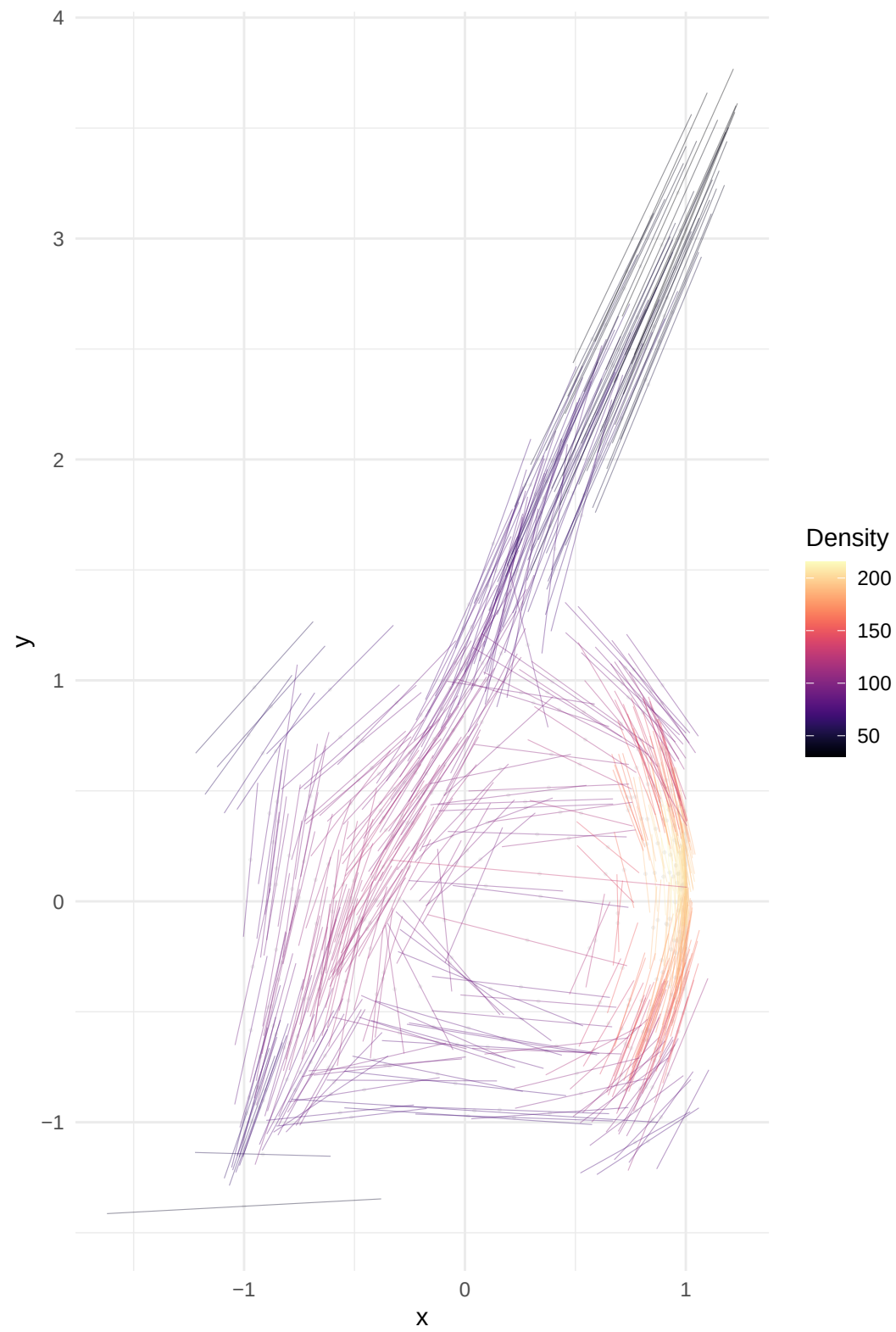
Visualizing local neighborhood size (radius) for every point.



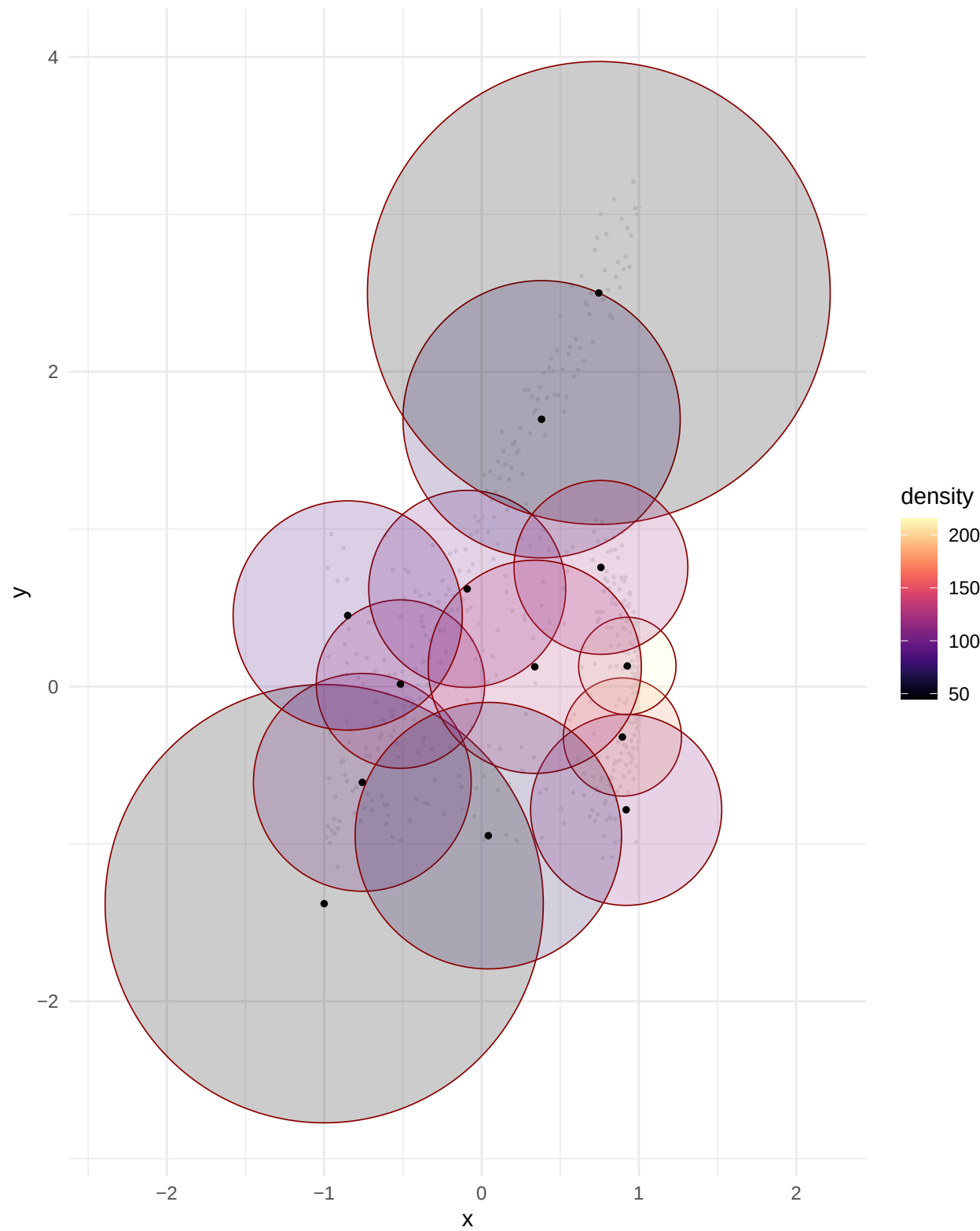
Point Density Distribution



Global Tangent Field (PC1)

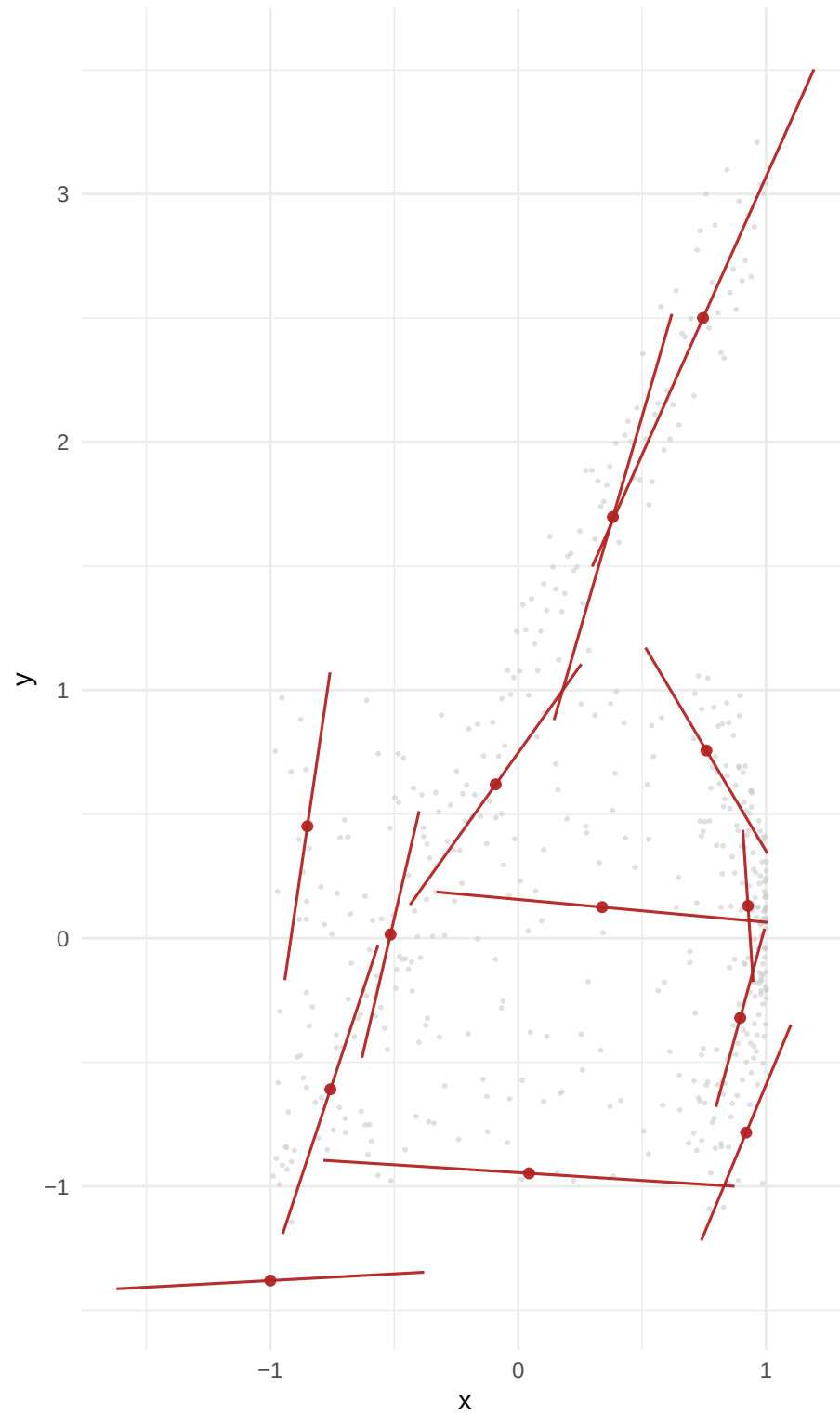


Atlas Selection (Anchors)



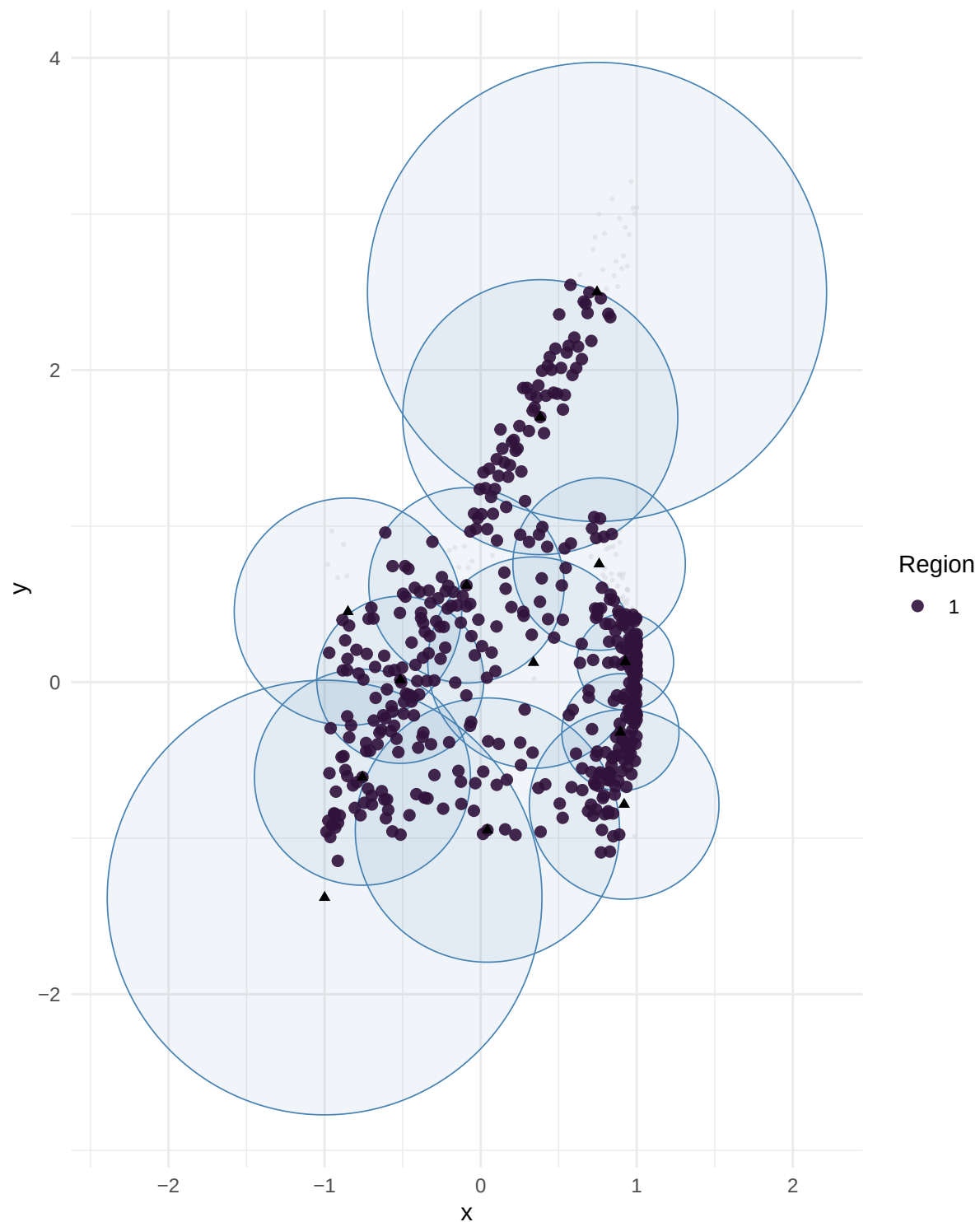
# Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



# Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Parameters: k\_min = 80, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01

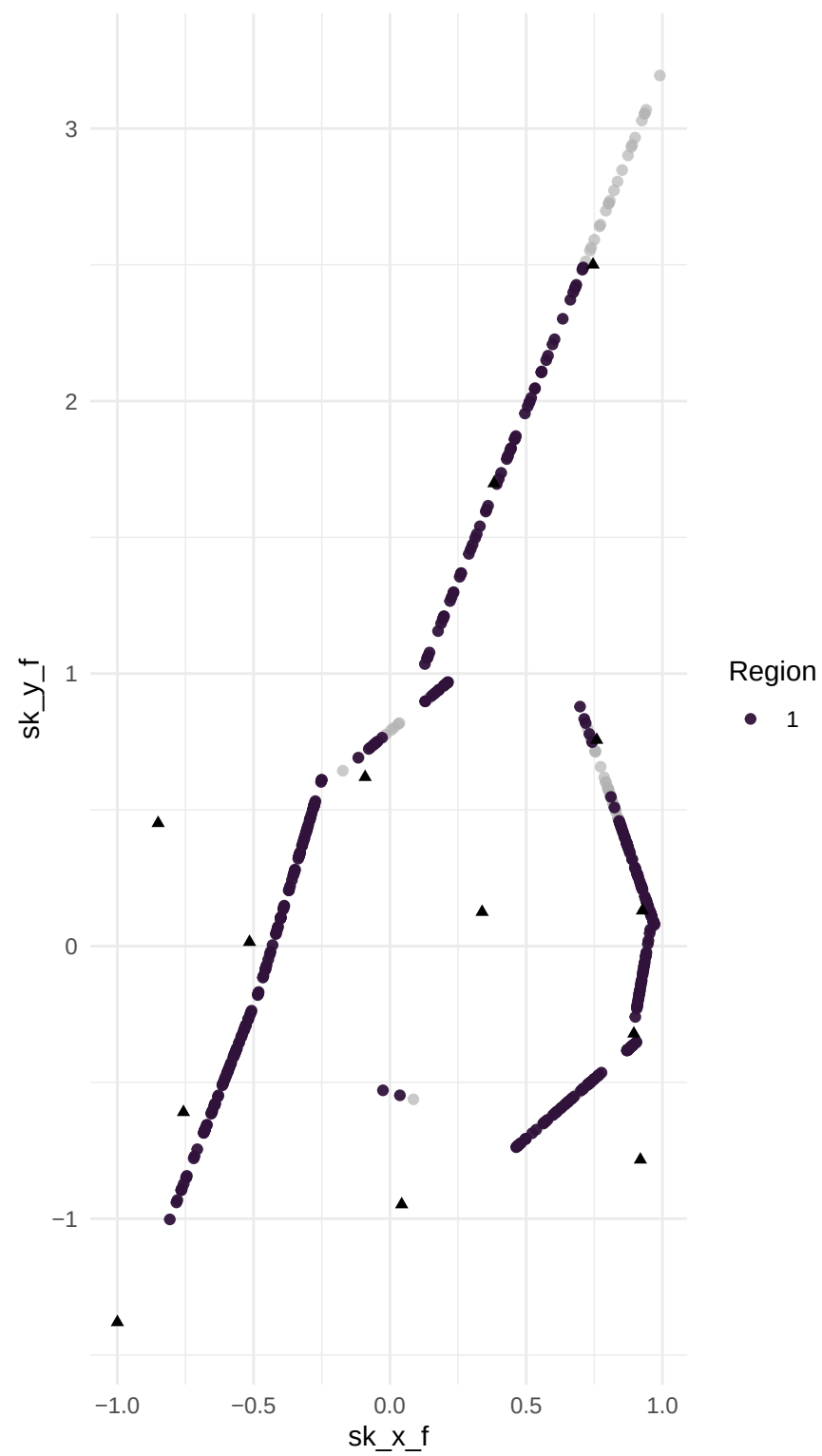
# Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors



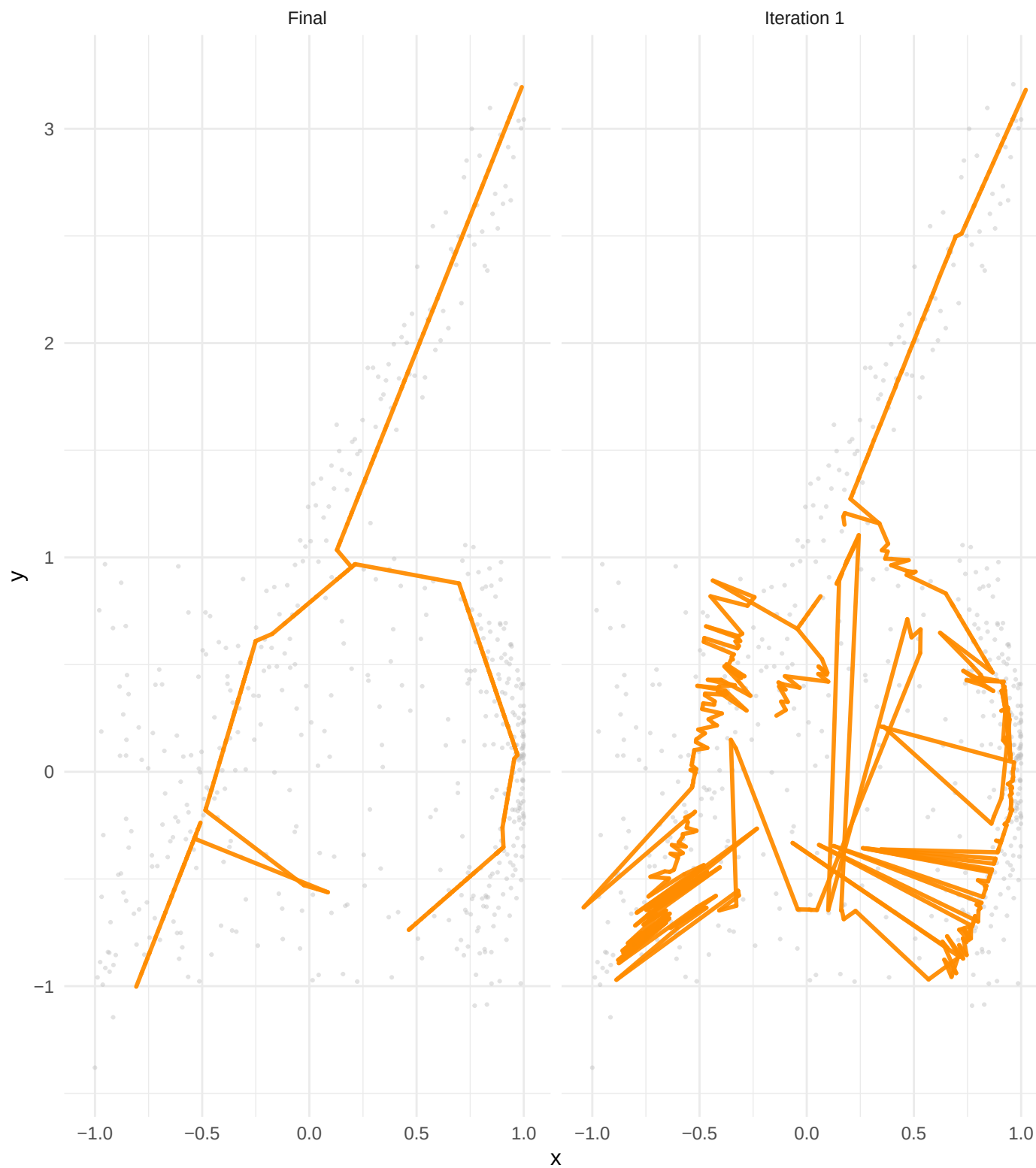
# Final Stitched Backbone (Converged)

Final stitched points

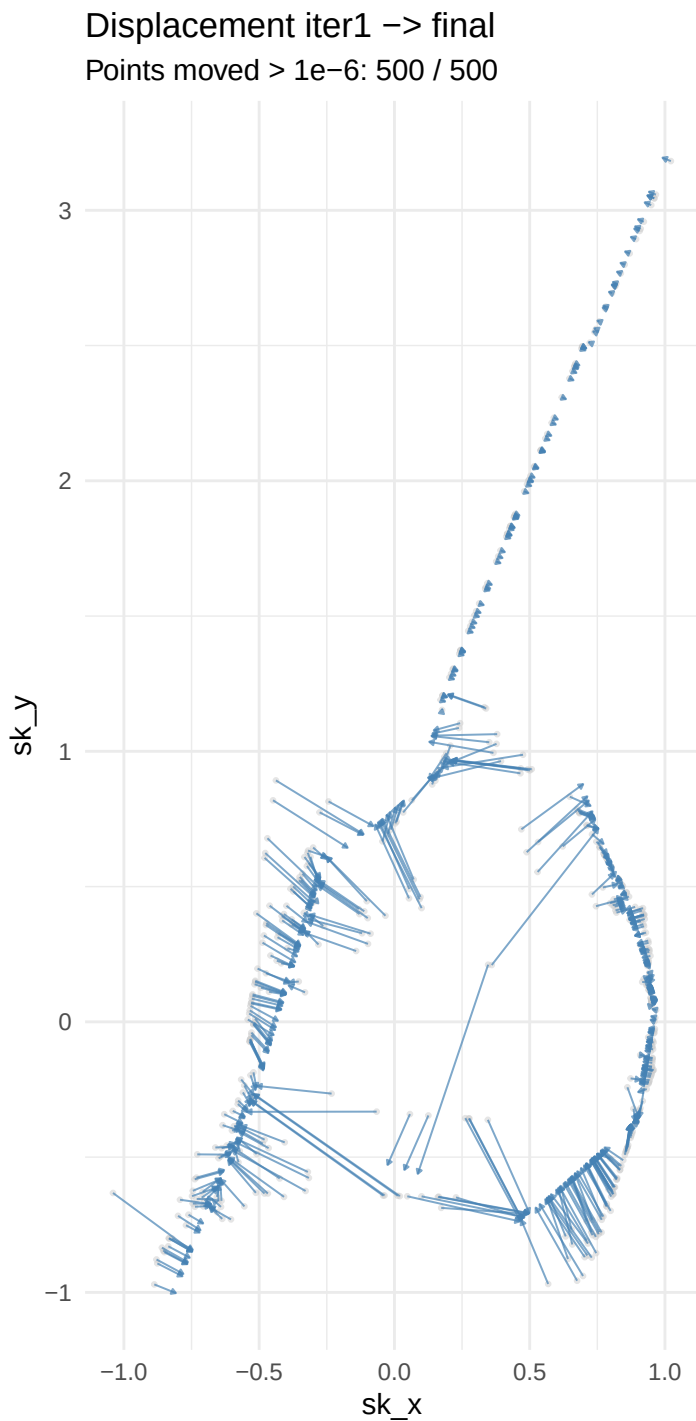
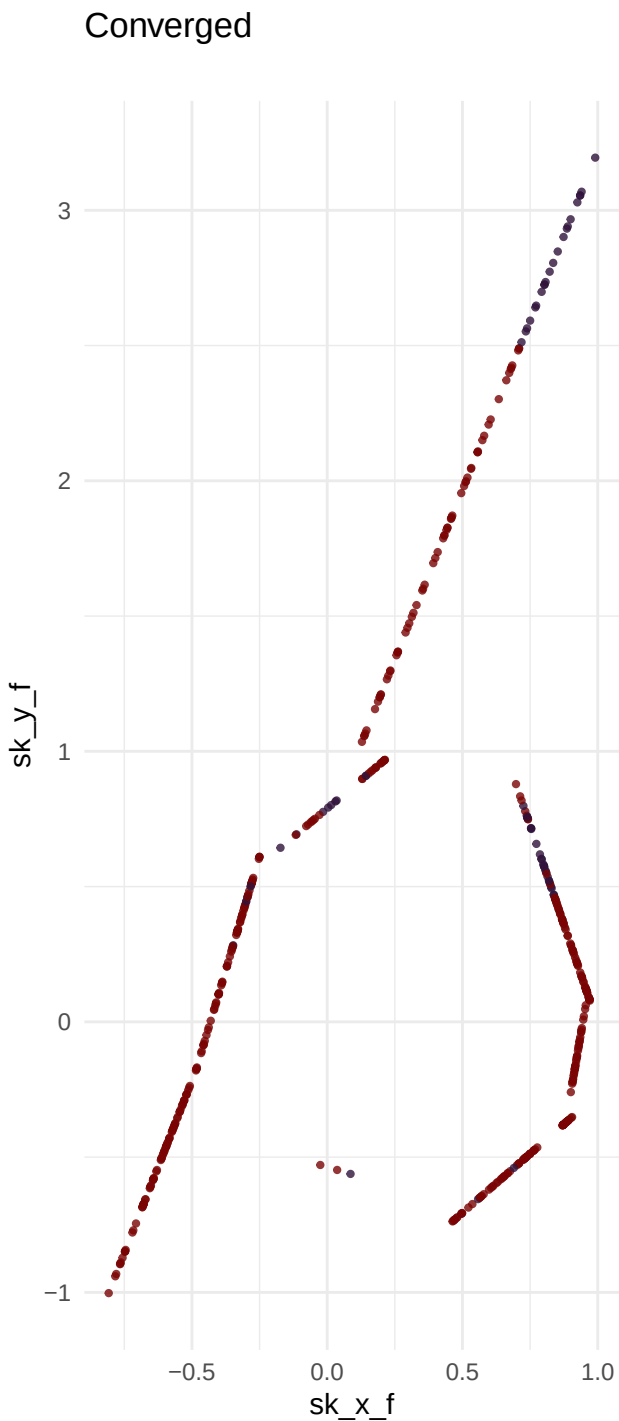
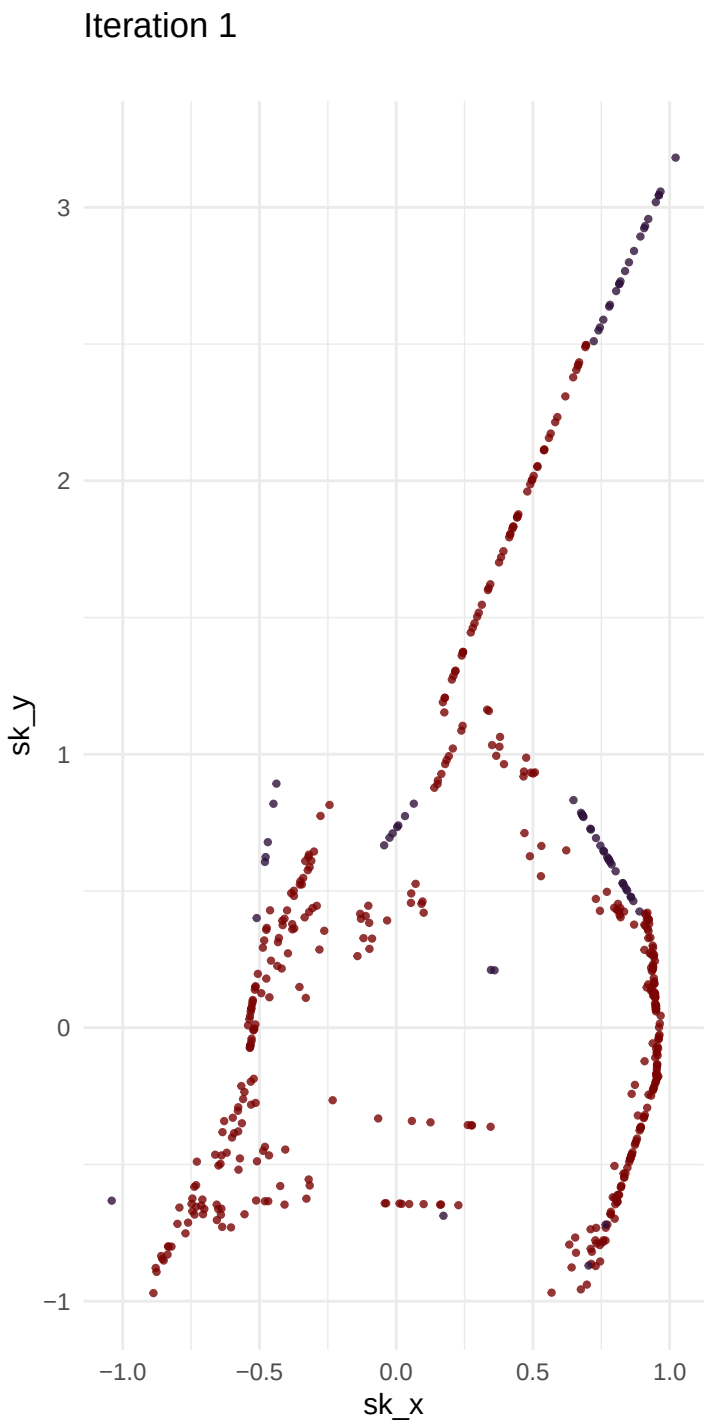




# Backbone Edges



Convergence: iteration 1 vs final



Parameters: k\_min = 80, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01